

Assignment 8: GLMs (Linear Regressions, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

NOTE: For this assignment, you can skip tests for any of the assumptions required to run the models.

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
knitr::opts_chunk$set(tidy = TRUE, tidy.opts = list(width.cutoff = 20))
library(agricolae)
library(tidyverse)
library(here)
getwd()

## [1] "C:/Users/24169/Desktop/e872/872_spring26"

lake_first <- read.csv(here(
  './Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv'
), stringsAsFactors = TRUE) %>%
  mutate(sampledate = mdy(sampledate))
#2
design <- theme_gray() + theme(plot.title = element_text(color = '#1C4087'),
                              plot.subtitle = element_text(color = '#5E45D9'),
                              axis.text = element_text(face = 'bold'),
                              panel.grid = element_blank(),
                              legend.background = element_rect(fill = '#FOFFFF'),
                              legend.position = 'bottom')
theme_set(design)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question:

- Answer

H0: The mean lake temperature recorded during July does not change with depth across all lakes.

Ha: The mean lake temperature recorded during July correlates with depth across all lakes.

4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:

- Only dates in July.
- Only the columns: lakename, year4, daynum, depth, temperature_C
- Only complete cases (i.e., remove NAs)

5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

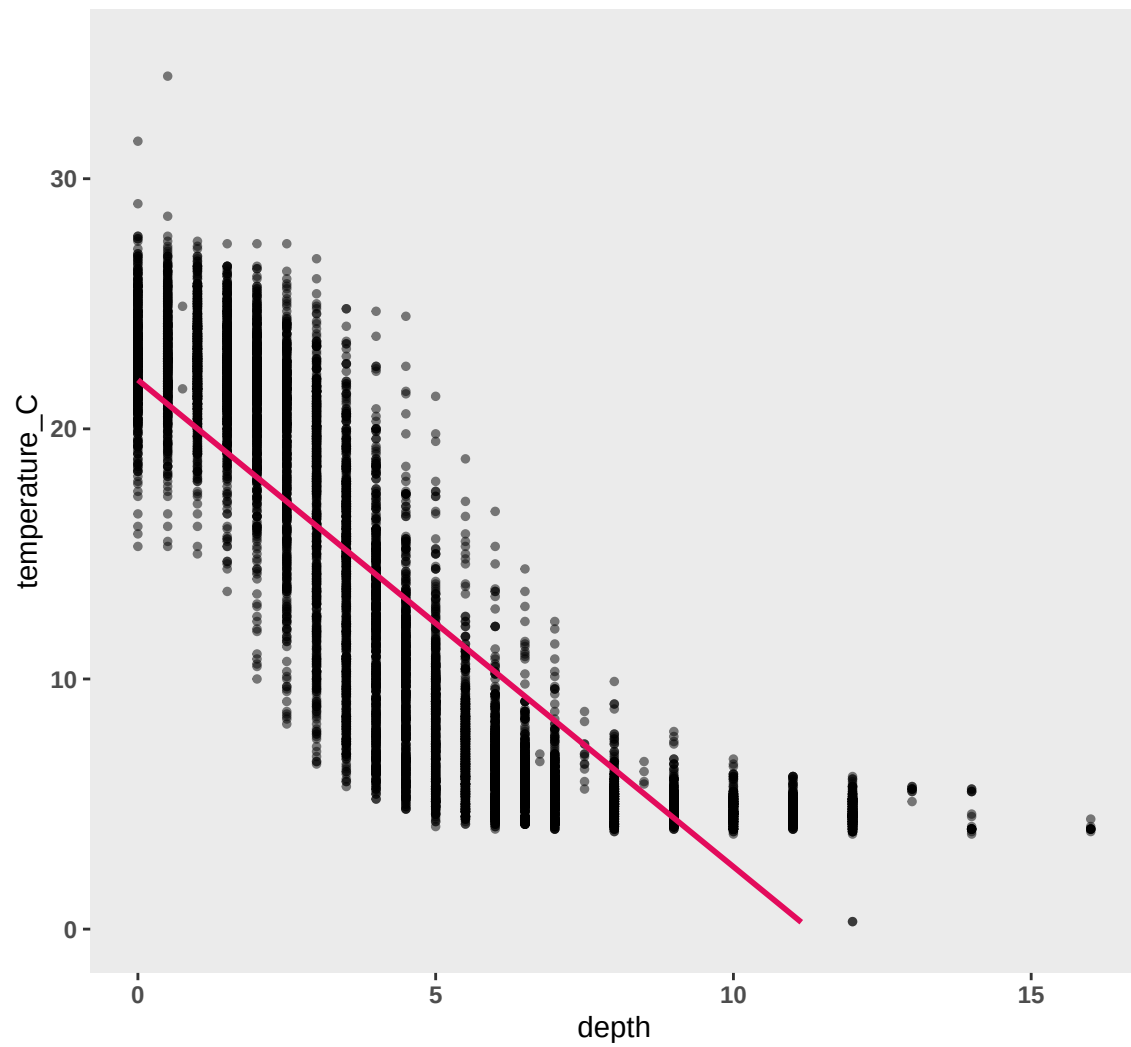
```

# 4
lake_second <- lake_first %>%
  filter(month(sampledate) ==
    7) %>%
  select(lakename,
    year4, daynum,
    depth, temperature_C) %>%
  drop_na()
# 5
p1 <- ggplot(data = lake_second,
  aes(x = depth, y = temperature_C)) +
  ylim(0, 35) + geom_point(size = 1,
  alpha = 0.5, color = "#000000") +
  geom_smooth(linewidth = 1,
    color = "#E30B5C",
    method = "lm",
    se = FALSE) +
  labs(title = "Relationship Between Lake Temperature And Lake Depth",
    subtitle = "Entire Observations")
p1

```

Relationship Between Lake Temperature And Lake Depth

Entire Observations



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

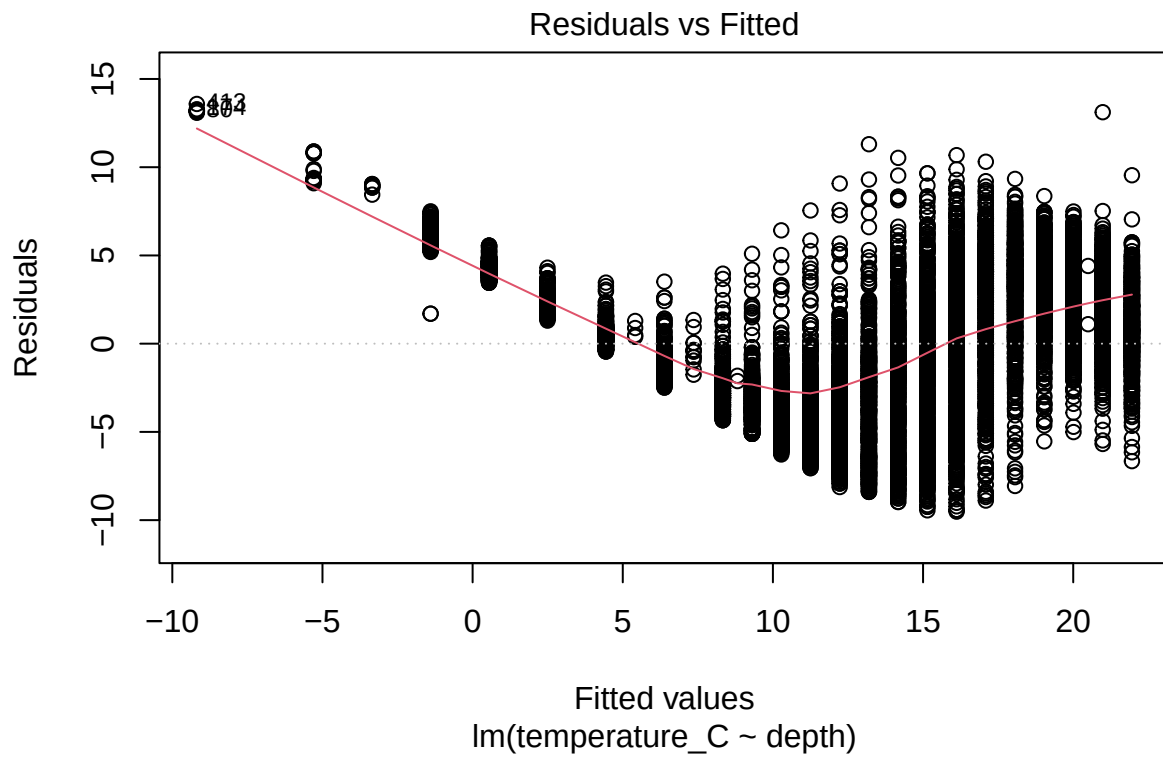
Answer: There might be a negative linear relationship between temperature and depth. In other words, if depth increases, temperature might decrease.

7. Perform a linear regression to test the relationship and display the results.

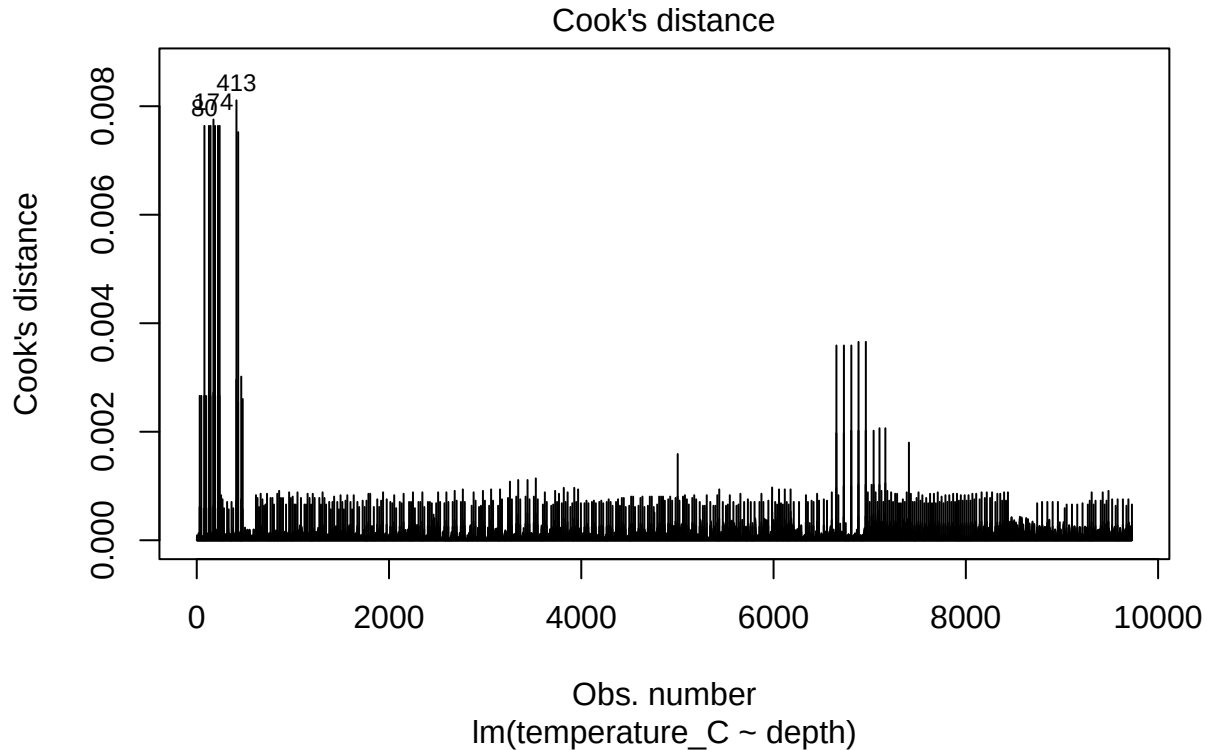
```
# 7
model_first <- lm(data = lake_second,
  temperature_C ~ depth)
summary(model_first)

##
## Call:
## lm(formula = temperature_C ~ depth, data = lake_second)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16

plot(model_first, 1)
```



```
plot(model_first, 4)
```



8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: If depth is zero meter, temperature will be 21.95597 Celsius degrees. If depth increases by one meter, temperature will decrease by 1.94621 Celsius degrees. Approximately 73.87% of variation in temperature can be interpreted by variation in depth. Given the very low p-value for coefficients and F-statistic, the null hypothesis is rejected. In other words, the relationship between temperature and depth might be significant.

Answer: number of all observation = 9728 | number of all explanatory variable = 1 | degrees of freedom for residual = $9728 - 1 - 1 = 9726$

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.

10. Run a multiple regression on the recommended set of variables.

```
# 9
model_second <- lm(data = lake_second,
  temperature_C ~ depth +
    daynum + year4)
step(model_second)

## Start:  AIC=26065.53
## temperature_C ~ depth + daynum + year4
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4    1         101 141788 26070
## - daynum   1         1237 142924 26148
## - depth    1      404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ depth + daynum + year4, data = lake_second)
##
## Coefficients:
## (Intercept)      depth      daynum      year4
##   -8.57556   -1.94644    0.03978    0.01134

model_third <- lm(data = lake_second,
  temperature_C ~ depth +
    daynum)
step(model_third)

## Start:  AIC=26070.49
## temperature_C ~ depth + daynum
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141788 26070
## - daynum   1         1241 143029 26153
## - depth    1      404384 546173 39188

##
## Call:
## lm(formula = temperature_C ~ depth + daynum, data = lake_second)
##
## Coefficients:
## (Intercept)      depth      daynum
##   14.08859   -1.94611    0.03984

model_fourth <- lm(data = lake_second,
  temperature_C ~ depth +
    year4)
step(model_fourth)
```

```
## Start: AIC=26148.11
## temperature_C ~ depth + year4
##
##           Df Sum of Sq    RSS   AIC
## <none>                142924 26148
## - year4  1             105 143029 26153
## - depth  1          404519 547443 39210

##
## Call:
## lm(formula = temperature_C ~ depth + year4, data = lake_second)
##
## Coefficients:
## (Intercept)      depth      year4
##    -1.10477    -1.94654     0.01154
```

```
# 10
summary(model_second)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + daynum + year4, data = lake_second)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## daynum        0.039780   0.004317   9.215 < 2e-16 ***
## year4         0.011345   0.004299   2.639  0.00833 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The model containing all three explanatory variables (year4, daynum, depth) should be supported. Any exclusion might result in higher AIC. Currently, approximately 74.11% of variation in temperature can be interpreted by variation in explanatory variables, which indicates a slight improvement.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
# 12
lake_third <- lake_second %>%
  group_by(lakename,
    year4, daynum) %>%
  summarise(temperature_C = mean(temperature_C))
model_fifth <- aov(data = lake_third,
  temperature_C ~ lakename)
summary(model_fifth)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8   1333   166.60   79.69 <2e-16 ***
## Residuals    524   1096     2.09
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_sixth <- lm(data = lake_third,
  temperature_C ~ lakename)
summary(model_sixth)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = lake_third)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.2754 -0.9386 -0.1271  0.7646  7.1552
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6870     0.3864  45.769 < 2e-16 ***
## lakenameCrampton Lake      -2.3176     0.5373  -4.313 1.92e-05 ***
## lakenameEast Long Lake     -7.4019     0.4382 -16.892 < 2e-16 ***
## lakenameHummingbird Lake   -6.8655     0.6178 -11.113 < 2e-16 ***
## lakenamePaul Lake         -3.8099     0.4041  -9.429 < 2e-16 ***
## lakenamePeter Lake        -4.2466     0.4041 -10.509 < 2e-16 ***
## lakenameTuesday Lake     -6.4968     0.4167 -15.590 < 2e-16 ***
## lakenameWard Lake         -3.2574     0.6408  -5.083 5.18e-07 ***
## lakenameWest Long Lake    -6.1034     0.4354 -14.019 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.446 on 524 degrees of freedom
## Multiple R-squared:  0.5489, Adjusted R-squared:  0.542
## F-statistic: 79.69 on 8 and 524 DF, p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: Given the very low p-value for F-statistic, at least one lake has an apparently different mean temperature.

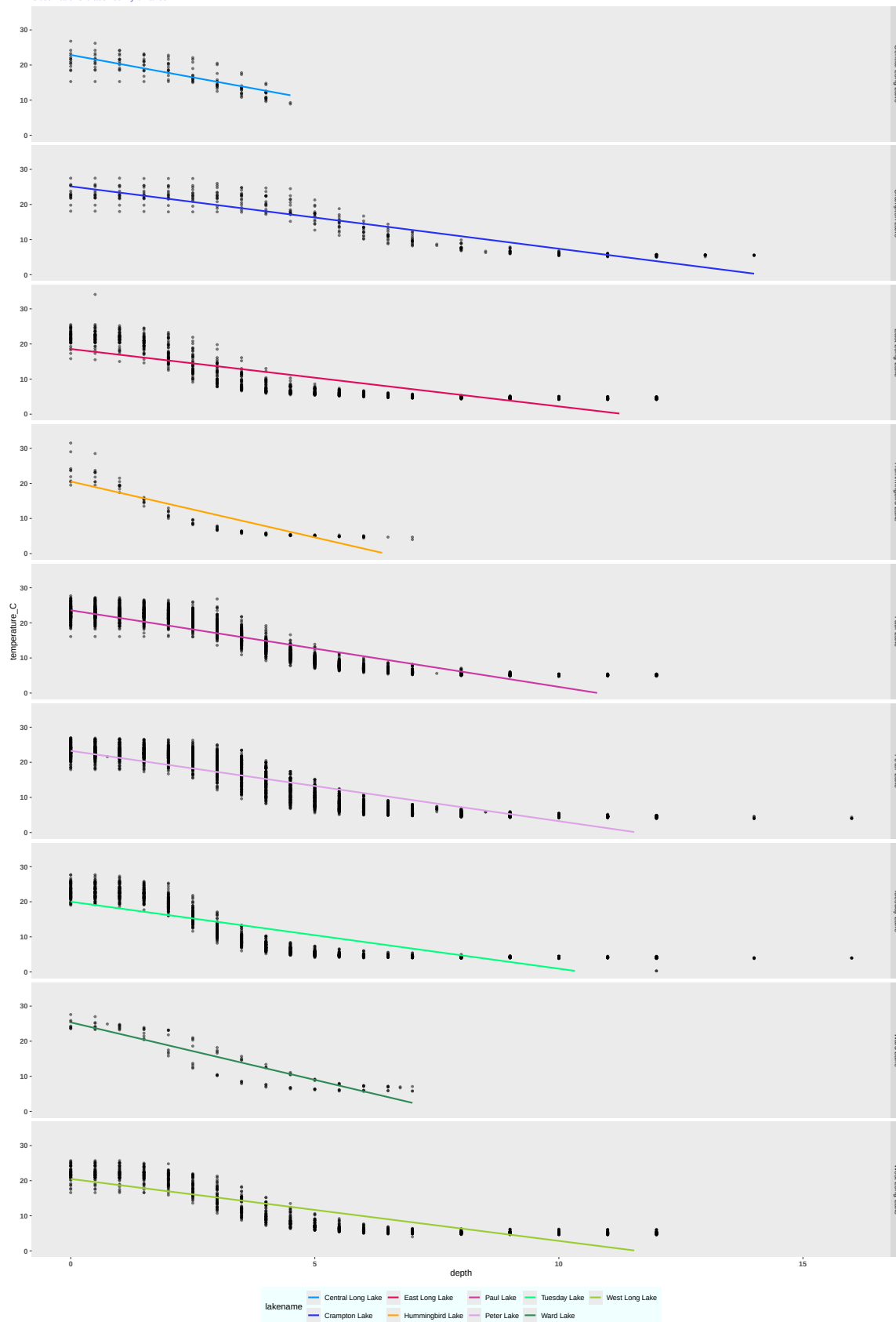
14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (`method = "lm"`, `se = FALSE`) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```

# 14
p2 <- ggplot(data = lake_second,
  aes(x = depth, y = temperature_C,
    color = lakename)) +
  ylim(0, 35) + geom_point(size = 1,
    alpha = 0.5, color = "#000000") +
  geom_smooth(linewidth = 1,
    method = "lm",
    se = FALSE) +
  scale_color_manual(values = c("#0096FF",
    "#2731F5", "#E30B5C",
    "#FFA500", "#CC39A5",
    "#DDA2E7", "#00FF7F",
    "#2E8B57", "#9ACD32")) +
  facet_grid(lakename ~
    .) + labs(title = "Relationship Between Lake Temperature And Lake Depth",
    subtitle = "Observations Classified By 9 Lakes")
p2

```

Relationship Between Lake Temperature And Lake Depth
Observations Classified By 9 Lakes



15. Use the Tukey's HSD test to determine which lakes have different means.

15

TukeyHSD(model_fifth)

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = lake_third)
##
## $lakename
##
```

	diff	lwr	upr	p adj
## Crampton Lake-Central Long Lake	-2.3175579	-3.99122964	-0.64388624	0.0006463
## East Long Lake-Central Long Lake	-7.4019316	-8.76679490	-6.03706840	0.0000000
## Hummingbird Lake-Central Long Lake	-6.8654742	-8.78971561	-4.94123270	0.0000000
## Paul Lake-Central Long Lake	-3.8099221	-5.06853792	-2.55130621	0.0000000
## Peter Lake-Central Long Lake	-4.2465818	-5.50519766	-2.98796595	0.0000000
## Tuesday Lake-Central Long Lake	-6.4967571	-7.79473610	-5.19877803	0.0000000
## Ward Lake-Central Long Lake	-3.2574254	-5.25352980	-1.26132101	0.0000182
## West Long Lake-Central Long Lake	-6.1034132	-7.45949914	-4.74732734	0.0000000
## East Long Lake-Crampton Lake	-5.0843737	-6.41338062	-3.75536680	0.0000000
## Hummingbird Lake-Crampton Lake	-4.5479162	-6.44689301	-2.64893942	0.0000000
## Paul Lake-Crampton Lake	-1.4923641	-2.71200406	-0.27272419	0.0048437
## Peter Lake-Crampton Lake	-1.9290239	-3.14866380	-0.70938393	0.0000394
## Tuesday Lake-Crampton Lake	-4.1791991	-5.43942024	-2.91897800	0.0000000
## Ward Lake-Crampton Lake	-0.9398675	-2.91162821	1.03189328	0.8624463
## West Long Lake-Crampton Lake	-3.7858553	-5.10584646	-2.46586414	0.0000000
## Hummingbird Lake-East Long Lake	0.5364575	-1.09687884	2.16979383	0.9835941
## Paul Lake-East Long Lake	3.5920096	2.85093181	4.33308736	0.0000000
## Peter Lake-East Long Lake	3.1553498	2.41427207	3.89642762	0.0000000
## Tuesday Lake-East Long Lake	0.9051746	0.09905302	1.71129615	0.0148765
## Ward Lake-East Long Lake	4.1445062	2.42709099	5.86192150	0.0000000
## West Long Lake-East Long Lake	1.2985184	0.40182930	2.19520753	0.0002729
## Paul Lake-Hummingbird Lake	3.0555521	1.50989696	4.60120722	0.0000001
## Peter Lake-Hummingbird Lake	2.6188923	1.07323722	4.16454748	0.0000068
## Tuesday Lake-Hummingbird Lake	0.3687171	-1.20915663	1.94659082	0.9983857
## Ward Lake-Hummingbird Lake	3.6080488	1.41958612	5.79651138	0.0000140
## West Long Lake-Hummingbird Lake	0.7620609	-0.86394796	2.38806980	0.8734535
## Peter Lake-Paul Lake	-0.4366597	-0.95671595	0.08339647	0.1826605
## Tuesday Lake-Paul Lake	-2.6868350	-3.29600999	-2.07766000	0.0000000
## Ward Lake-Paul Lake	0.5524967	-1.08175465	2.18674797	0.9802906
## West Long Lake-Paul Lake	-2.2934912	-3.01827635	-1.56870599	0.0000000
## Tuesday Lake-Peter Lake	-2.2501753	-2.85935025	-1.64100026	0.0000000
## Ward Lake-Peter Lake	0.9891564	-0.64509491	2.62340771	0.6242681
## West Long Lake-Peter Lake	-1.8568314	-2.58161661	-1.13204625	0.0000000
## Ward Lake-Tuesday Lake	3.2393317	1.57457550	4.90408782	0.0000001
## West Long Lake-Tuesday Lake	0.3933438	-0.39782573	1.18451338	0.8317994
## West Long Lake-Ward Lake	-2.8459878	-4.55643586	-1.13553981	0.0000111

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Paul Lake and Ward Lake might have the same mean temperature as Peter Lake. In

addition, Central Long Lake has a mean temperature that is statistically distinct from all other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: We might consider a two-sample T-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
# 18
lake_fourth <- lake_second %>%
  filter(lakename ==
         "Crampton Lake" |
         lakename == "Ward Lake")
examine <- t.test(lake_fourth$temperature ~
  lake_fourth$lakename)
examine

##
## Welch Two Sample t-test
##
## data: lake_fourth$temperature by lake_fourth$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.35189                14.45862
```

Answer: Since the p-value is 0.2649, we cannot reject the null hypothesis. The mean temperature at Ward Lake might be equal to that at Crampton Lake. Our result here matches the Tukey's HSD test, which generates a p-value of 0.8624463.